## A Dataset

This work uses benchmark Super-Natural Instructions (Wang et al., 2022b) which includes 1,616 diverse NLP datasets covering 76 distinct task types. We select 15 NLG task types and list all datasets included in each NLG task in Tab. A2.

## B Results and Analysis

**Separate Metric as Continuous *Regression* Label** In addition to the comprehensive integration of all metrics (i.e. NLI, Rouge-L, Questeval) described in § 3.2, we analyze our internal state-based method's performances when treating each metric as the label, separately.

We consider three forms of "golden score" for each metric. First, the absolute values, which serve as the target for regression, with higher scores indicating fewer hallucinations. We consider the probability of entailment as the absolute value of the NLI metric. Second, we standardize these absolute values using the minimum and maximum values from the training dataset to obtain normalized "golden scores". Third, we use the relative rankings of these scores within the training dataset as an alternative regression target.

For the regression task with continuous score predicted, we utilize **Root Mean Squared Error (RMSE)** to measure the average difference between the values predicted by our estimator and the actual values.

As shown in Fig. A1, our method's prediction performance varies across the form of the "golden score". For each metric, the RMSE is the smallest when predicting absolute value, which indicates that the hidden state performs best in predicting the absolute value of the metric. Conversely, the highest RMSE occurs when the model attempts to predict the relative rankings, implying that predicting the precise ordering of the metrics is more challenging for the hidden state representation.

**Estimator Backbone** Instead of Llama MLP, we employ a standard MLP as the backbone of the estimator. The results in Tab. A1 demonstrate that Llama MLP outperforms the standard MLP.

## C Implementation Details

The input dimension of our classifier is 4096 and the hidden dimension is 11008, which are aligned with Llama2-7B. We train our classifier with the following settings and hyper-parameters: the epoch is 10, the batch size is 128, the learning rate is 1e-5, and the AdamW optimizer has a linear scheduler. Our model is trained on 1 NVIDIA A800 GPU.

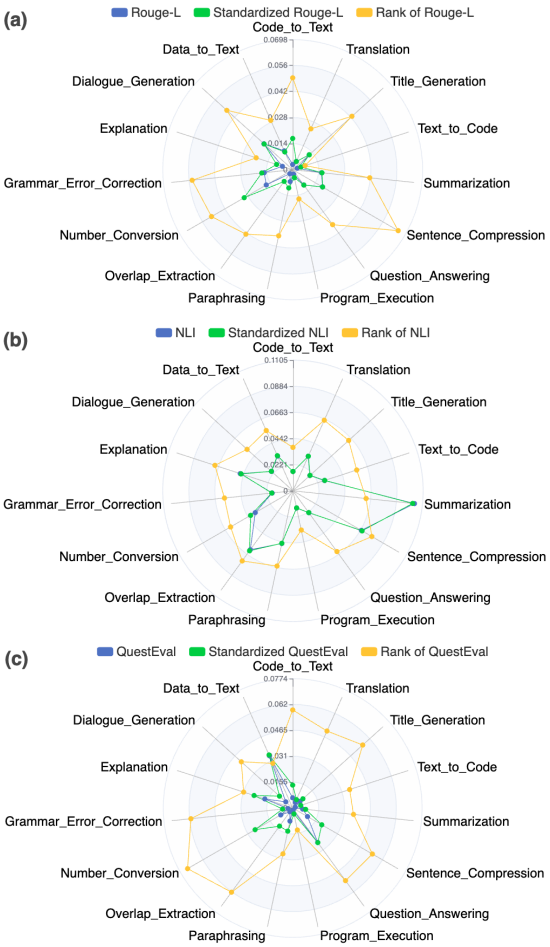



Figure A1: RMSE Scores of Internal State-based Estimator with Labels: (a) Rouge-L (b) NLI (c) QuestEval

| Task | Internal State | F1 | ACC |
|---|---|---|---|
| Dialogue | LlamaMLP | **74.33** | **74.22** |
| | MLP | 70.22 | 71.12 |
| QA | LlamaMLP | **82.37** | **82.55** |
| | MLP | 81.57 | 81.84 |
| Summarization | LlamaMLP | **88.08** | **88.95** |
| | MLP | 87.59 | 87.59 |
| Translation | LlamaMLP | **76.90** | **76.90** |
| | MLP | 74.23 | 74.90 |

Table A1: Automatic Evaluation Results for Different Classifier Backbone

## D AI Assistants Using

In this paper, we use ChatGPT to improve the writing at the grammar level.

| Task | No. | Dataset |
|---|---|---|
| Code to Text | 4 | Task 110: logic2text sentence generation, Task 129: scan long text generation action command short, Task 127: scan long text generation action command all, Task 131: scan long text generation action command long |
| Data to Text | 9 | Task 1728: web nlg data to text, Task 1598: nyc long text generation, Task 1631: openpi answer generation, Task 677: ollie sentence answer generation, Task 957: e2e nlg text generation generate, Task 760: msr sqa long text generation, Task 1407: dart question generation, Task 102: commongen sentence generation, Task 1409: dart text generation |
| Dialogue Generation | 13 | Task 574: air dialogue sentence generation, Task 361: spolin yesand prompt response classification, Task 576: curiosity dialogs answer generation, Task 1603: smcalflow sentence generation, Task 1714: convai3 sentence generation, Task 1730: personachat choose next, Task 565: circa answer generation, Task 611: mutual multi turn dialogue, Task 1729: personachat generate next, Task 1600: smcalflow sentence generation, Task 639: multi woz user utterance generation, Task 1590: diplomacy text generation, Task 360: spolin yesand response generation |
| Explanation | 6 | Task 295: semeval 2020 Task 4: commonsense reasoning, Task 192: hotpotqa sentence generation, Task 593: sciq explanation generation, Task 1369: healthfact sentence generation, Task 223: quartz explanation generation, Task 134: winowhy reason generation |
| Grammar Error Correction | 2 | Task 1415: youtube caption corrections grammar correction, Task 1557: jfleg answer generation |
| Number Conversion | 2 | Task 1703: ljspeech textmodification, Task 1704: ljspeech textmodification |
| Overlap Extraction | 2 | Task 039: qasc find overlapping words, Task 281: points of correspondence |
| Paraphrasing | 12 | Task 776: pawsx japanese text modification, Task 045: miscellaneous sentence paraphrasing, Task 770: pawsx english text modification, Task 771: pawsx korean text modification, Task 774: pawsx german text modification, Task 177: para-nmt paraphrasing, Task 466: parsinlu qqp text modification, Task 775: pawsx chinese text modification, Task 1614: sick text modify, Task 773: pawsx spanish text modification, Task 132: dais text modification, Task 772: pawsx french text modification |
| Program Execution | 90 | Task 113: count frequency of letter, Task 1151: swap max min, Task 509: collate of all alphabetical and numerical elements in list separately, Task 100: concatenate all elements from index i to j, Task 096: conala list index subtraction, Task 365: synthetic remove vowels, Task 622: replace alphabets in a list by their position in english alphabet, Task 852: synthetic multiply odds, Task 1088: array of products, Task 1405: find median, Task 637: extract and sort unique digits in a list, Task 1446: farthest integers, Task 506: position of all alphabetical elements in list, Task 378: reverse words of given length, Task 093: conala normalize lists, Task 1404: date conversion, Task 097: conala remove duplicates, Task 372: synthetic palindrome numbers, Task 755: find longest substring and replace its sorted lowercase version in both lists, Task 636: extract and sort unique alphabets in a list, Task 267: concatenate and reverse all elements from index i to j, Task 162: count words starting with letter, Task 159: check frequency of words in sentence pair, Task 208: combinations of list, Task 1316: remove duplicates string, Task 504: count all alphabetical elements in list, Task 079: conala concat strings, Task 158: count frequency of words, Task 507: position of all numerical elements in list, Task 374: synthetic pos or neg calculation, Task 1087: two number sum, Task 163: count words ending with letter, Task 756: find longert substring and return all unique alphabets in it, Task 101: reverse and concatenate all elements from index i to j, Task 1551: every ith element from kth element, Task 606: sum of all numbers in list between positions i and j, Task 368: synthetic even or odd calculation, Task 1150: delete max min, Task 851: synthetic multiply evens, Task 377: remove words of given length, Task 063: first i elements, Task 064: all elements except first i, Task 245: check presence in set intersection, Task 161: count words containing letter, Task 605: find the longest common subsequence in two lists, Task 850: synthetic longest palindrome, Task 157: count vowels and consonants, Task 373: synthetic round tens place, Task 206: collatz conjecture, Task 1443: string to number, Task 123: conala sort dictionary, Task 244: count elements in set union, Task 499: extract and add all numbers from list, Task 124: conala pair averages, Task 1444: round power of two, Task 099: reverse elements between index i and j, Task 1089: check monotonic array, Task 1188: count max freq char, Task 125: conala pair differences, Task 488: extract all alphabetical elements from list in order, Task 1542: every ith element from starting, Task 1194: kth largest element, Task 371: synthetic product of list, Task 1406: kth smallest element, Task 095: conala max absolute value, Task 1315: find range array, Task 243: count elements in set intersection, Task 1331: reverse array, Task 062: bigbench repeat copy logic, Task 122: conala list index addition, Task 091: all elements from index i to j, Task 369: synthetic remove odds, Task 497: extract all numbers from list in order, Task 505: count all numerical elements in list, Task 205: remove even elements, Task 1189: check char in string, Task 1445: closest integers, Task 094: conala calculate mean, Task 160: replace letter in a sentence, Task 1148: maximum ascii value, Task 098: conala list intersection, Task 078: all elements except last i, Task 523: find if numbers or alphabets are more in list, Task 370: synthetic remove divisible by 3, Task 367: synthetic remove floats, Task 1190: add integer to list, Task 376: reverse order of words, Task 600: find the longest common substring in two strings, Task 207: max element lists, Task 366: synthetic return primes |

Table A1 – continued from previous page

| Task | No. | Dataset |
|---|---|---|
| Question Answering | 207 | Task 837: viquiquad answer generation, Task 701: mmmlu answer generation high school computer science, Task 1399: obqa answer generation, Task 075: squad1.1 answer generation, Task 724: mmmlu answer generation moral scenarios, Task 666: mmmlu answer generation astronomy, Task 742: lhoestq answer generation frequency, Task 1438: doqa cooking answer generation, Task 863: asdiv multiop question answering, Task 864: asdiv singleop question answering, Task 058: multirc question answering, Task 669: ambigqa answer generation, Task 704: mmmlu answer generation high school government and politics, Task 728: mmmlu answer generation professional accounting, Task 740: lhoestq answer generation quantity, Task 1293: kilt tasks hotpotqa question answering, Task 849: pubmedqa answer generation, Task 1424: mathqa probability, Task 1625: disfl qa asnwer generation, Task 858: inquisitive span detection, Task 723: mmmlu answer generation moral disputes, Task 083: babi t1 single supporting fact answer generation, Task 118: semeval 2019 Task 10: open vocabulary mathematical answer generation, Task 582: naturalquestion answer generation, Task 237: iirc answer from subtext answer generation, Task 714: mmmlu answer generation human sexuality, Task 444: com qa question paraphrases answer generation, Task 720: mmmlu answer generation marketing, Task 332: tellmewhy answer generation, Task 119: semeval 2019 Task 10: geometric mathematical answer generation, Task 310: race classification, Task 1132: xcsr ur commonsense mc classification, Task 702: mmmlu answer generation high school european history, Task 710: mmmlu answer generation high school statistics, Task 870: msmarco answer generation, Task 047: miscellaneous answering science questions, Task 711: mmmlu answer generation high school us history, Task 1286: openbookqa question answering, Task 598: cuad answer generation, Task 685: mmmlu answer generation clinical knowledge, Task 084: babi t1 single supporting fact identify relevant fact, Task 1420: mathqa general, Task 1520: qa srl answer generation, Task 868: mawps singleop question answering, Task 768: qed text span selection, Task 061: ropes answer generation, Task 041: qasc answer generation, Task 144: subjqa question answering, Task 1570: cmrc2018 answer generation, Task 1610: xquad es answer generation, Task 164: mcscript question answering text, Task 703: mmmlu answer generation high school geography, Task 705: mmmlu answer generation high school macroeconomics, Task 1131: xcsr es commonsense mc classification, Task 1130: xcsr vi commonsense mc classification, Task 750: aqua multiple choice answering, Task 473: parsinlu mc classification, Task 385: socialiqa incorrect answer generation, Task 691: mmmlu answer generation college physics, Task 719: mmmlu answer generation management, Task 1327: qa zre answer generation from question, Task 715: mmmlu answer generation international law, Task 737: mmmlu answer generation world religions, Task 010: mctaco answer generation event ordering, Task 741: lhoestq answer generation place, Task 028: drop answer generation, Task 730: mmmlu answer generation professional medicine, Task 491: mwsc answer generation, Task 716: mmmlu answer generation jurisprudence, Task 732: mmmlu answer generation public relations, Task 735: mmmlu answer generation us foreign policy, Task 898: freebase qa answer generation, Task 887: quail answer generation, Task 024: cosmosqa answer generation, Task 1140: xcsr pl commonsense mc classification, Task 225: english language answer generation, Task 1608: xquad en answer generation, Task 170: hotpotqa answer generation, Task 667: mmmlu answer generation business ethics, Task 699: mmmlu answer generation high school biology, Task 595: mocha answer generation, Task 751: svamp subtraction question answering, Task 1656: gooaq answer generation, Task 1431: head qa answer generation, Task 1296: wiki hop question answering, Task 490: mwsc options generation, Task 867: mawps multiop question answering, Task 865: mawps addsub question answering, Task 1133: xcsr nl commonsense mc classification, Task 1422: mathqa physics, Task 1135: xcsr en commonsense mc classification, Task 054: multirc write correct answer, Task 1661: super glue classification, Task 708: mmmlu answer generation high school physics, Task 1726: mathqa correct answer generation, Task 664: mmmlu answer generation abstract algebra, Task 1412: web questions question answering, Task 002: quoref answer generation, Task 752: svamp multiplication question answering, Task 1297: qasc question answering, Task 692: mmmlu answer generation computer security, Task 1136: xcsr fr commonsense mc classification, Task 727: mmmlu answer generation prehistory, Task 725: mmmlu answer generation nutrition, Task 104: semeval 2019 Task 10: closed vocabulary mathematical answer generation, Task 694: mmmlu answer generation econometrics, Task 820: protoqa answer generation, Task 700: mmmlu answer generation high school chemistry, Task 390: torque text span selection, Task 1421: mathqa other, Task 918: coqa answer generation, Task 309: race answer generation, Task 247: dream answer generation, Task 695: mmmlu answer generation electrical engineering, Task 230: iirc passage classification, Task 712: mmmlu answer generation high school world history, Task 731: mmmlu answer generation professional psychology, Task 596: mocha question generation, Task 698: mmmlu answer generation global facts, Task 718: mmmlu answer generation machine learning, Task 395: persianqa answer generation, Task 597: cuad answer generation, Task 339: record answer generation, Task 835: mathdataset answer generation, Task 238: iirc answer from passage answer generation, Task 228: arc answer generation easy, Task 380: boolq yes no question, Task 152: tomqa find location easy noise, Task 754: svamp common-division question answering, Task 713: mmmlu answer generation human aging, Task 665: mmmlu answer generation anatomy, Task 706: mmmlu answer generation high school mathematics, Task 697: mmmlu answer generation formal logic, Task 753: svamp addition question answering, Task 1727: wiqa what is the effect, Task 1139: xcsr ru commonsense mc classification, Task 1134: xcsr hi commonsense mc classification, Task 344: hybridqa answer generation, Task 165: mcscript question answering commonsense, Task 1145: xcsr jap commonsense mc classification, Task 1295: adversarial qa question answering, Task 239: tweetqa answer generation, Task 1382: quarel write correct answer... |

Table A1 – continued from previous page

| Task | No. | Dataset |
|---|---|---|
| Translation | 394 | Task 808: pawsx chinese korean translation, Task 254: spl translation fi en, Task 1111: ted translation he it, Task 988: pib translation oriya english, Task 650: opus100 ar en translation, Task 763: emea es lt translation, Task 1648: opus books en-sv translation, Task 1263: ted translation pl fa, Task 1020: pib translation telugu oriya, Task 913: bianet translation, Task 1060: pib translation urdu malayalam, Task 1676: xquad-ca translation, Task 1098: ted translation ja fa, Task 984: pib translation marathi gujarati, Task 1086: pib translation marathi english, Task 789: pawsx french english translation, Task 1110: ted translation he gl, Task 1689: qed amara translation, Task 787: pawsx korean chinese translation, Task 1071: pib translation malayalam marathi, Task 548: alt translation en ch, Task 1373: newscomm translation, Task 1023: pib translation english hindi, Task 1271: ted translation fa it, Task 1274: ted translation pt en, Task 552: alt translation en bu, Task 1040: pib translation punjabi oriya, Task 1323: open subtitles hi en translation, Task 1058: pib translation urdu english, Task 1105: ted translation ar gl, Task 1353: hind encorp translation en hi, Task 1085: pib translation english marathi, Task 1103: ted translation es fa, Task 784: pawsx korean french translation, Task 811: pawsx chinese german translation, Task 1365: opustedtalks translation, Task 1278: ted translation pt he, Task 1115: alt ja id translation, Task 538: alt translation bu en, Task 786: pawsx korean german translation, Task 805: pawsx german chinese translation, Task 1692: qed amara translation, Task 1690: qed amara translation, Task 655: bible en fa translation, Task 1256: ted translation pl en, Task 977: pib translation oriya urdu, Task 841: para pdt de en translation, Task 996: pib translation english bengali, Task 531: europarl es en translation, Task 452: opus paracrawl en ig translation, Task 1250: ted translation it ar, Task 644: refresd translation, Task 1248: ted translation it ja, Task 1034: pib translation hindi gujarati, Task 1225: ted translation ja he, Task 997: pib translation bengali oriya, Task 1127: alt ja th translation, Task 783: pawsx korean english translation, Task 1031: pib translation bengali telugu, Task 560: alt translation en entk, Task 1000: pib translation tamil malayalam, Task 252: spl translation en tr, Task 1650: opus books en-fi translation, Task 654: bible fa en translation, Task 802: pawsx german korean translation, Task 1025: pib translation bengali punjabi, Task 262: spl translation ja en, Task 785: pawsx korean spanish translation, Task 530: europarl en es translation, Task 1232: ted translation ar es, Task 799: pawsx spanish chinese translation, Task 1119: alt fil ja translation, Task 260: spl translation zh en, Task 1686: menyo20k translation, Task 448: opus paracrawl en tl translation, Task 994: pib translation tamil hindi, Task 1065: pib translation punjabi telugu, Task 557: alt translation en ba, Task 1072: pib translation marathi malayalam, Task 535: alt translation ch en, Task 762: emea fr sk translation, Task 1024: pib translation hindi english, Task 914: bianet translation, Task 779: pawsx english spanish translation, Task 547: alt translation entk en, Task 1128: alt th ja translation, Task 537: alt translation th en, Task 1277: ted translation pt ar, Task 1124: alt ja lo translation, Task 1514: flores translation entone, Task 435: alt en ja translation, Task 425: hindienglish corpora en hi translation, Task 1371: newscomm translation, Task 818: pawsx japanese chinese translation, Task 873: opus xhosanavy translation xhosa eng, Task 1240: ted translation gl es, Task 553: alt translation en ma, Task 1351: opus100 translation gu en, Task 999: pib translation malayalam tamil, Task 438: eng guj parallel corpus en gu translation, Task 541: alt translation kh en, Task 1329: open subtitles en hi translation, Task 1102: ted translation es pl, Task 661: mizan en fa translation, Task 1259: ted translation pl ar, Task 424: hindienglish corpora hi en translation, Task 793: pawsx french chinese translation, Task 1005: pib translation malayalam punjabi, Task 1262: ted translation pl it, Task 1367: opustedtalks translation, Task 117: spl translation en de, Task 1237: ted translation he ar, Task 1122: alt khm ja translation, Task 1230: ted translation ar en, Task 790: pawsx french korean translation, Task 433: alt hi en translation, Task 253: spl translation en zh, Task 1037: pib translation telugu urdu, Task 840: para pdt en es translation, Task 982: pib translation tamil bengali, Task 1009: pib translation bengali hindi, Task 1062: pib translation marathi bengali, Task 1218: ted translation en ja, Task 1113: ted translation he fa, Task 1691: qed amara translation, Task 1276: ted translation pt es, Task 1108: ted translation ar fa, Task 1070: pib translation urdu bengali, Task 1244: ted translation gl pl, Task 1239: ted translation gl ja, Task 1055: pib translation marathi oriya, Task 794: pawsx french japanese translation, Task 1004: pib translation malayalam bengali, Task 1049: pib translation malayalam telugu, Task 989: pib translation marathi urdu, Task 450: opus paracrawl so en translation, Task 815: pawsx japanese french translation, Task 1066: pib translation telugu punjabi, Task 777: pawsx english korean translation, Task 542: alt translation ja en, Task 830: poleval2019 mt translation, Task 1655: mkb translation, Task 313: europarl en sv translation, Task 1044: pib translation punjabi gujarati, Task 1038: pib translation urdu telugu, Task 1057: pib translation english urdu, Task 1047: pib translation english telugu, Task 1258: ted translation pl es, Task 1001: pib translation gujarati urdu, Task 1063: pib translation gujarati tamil, Task 1649: opus books en-no translation, Task 1282: ted translation pt fa, Task 983: pib translation gujarati marathi, Task 261: spl translation es en, Task 439: eng guj parallel corpus gu en translation, Task 795: pawsx spanish english translation, Task 1046: pib translation telugu hindi, Task 1233: ted translation ar he, Task 1112: ted translation he pl, Task 663: global voices en fa translation, Task 662: global voices fa en translation, Task 1376: newscomm translation, Task 258: spl translation fa en, Task 1029: pib translation marathi punjabi, Task 986: pib translation oriya hindi, Task 1067: pib translation bengali gujarati, Task 604: flores translation entosn, Task 1224: ted translation ja ar, Task 250: spl translation en ar, Task 1242: ted translation gl he, Task 559: alt translation en fi, Task 1015: pib translation punjabi tamil, Task 259: spl translation tr en, Task 1269: ted translation fa he, Task 807: pawsx chinese english translation, Task 809: pawsx chinese french translation, Task 995: pib translation bengali english, Task 1093: ted translation en fa, Task 174: spl translation en ja, Task 1036: pib translation urdu tamil, Task 1245: ted translation gl fa, Task 1081: pib translation hindi marathi, Task 1249: ted translation it es, Task 842: para pdt cs en translation, Task 1006: pib translation punjabi malayalam, Task 1091: ted translation en it, Task 1022: pib translation malayalam english, Task 656: quran en fa translation... |